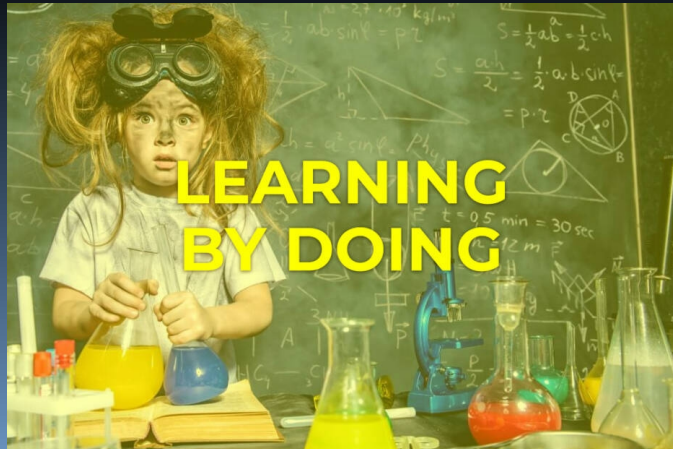




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Snake Game



For the things we have to learn before we can do them, we learn by doing them

Aristotle, The Nicomachean Ethics

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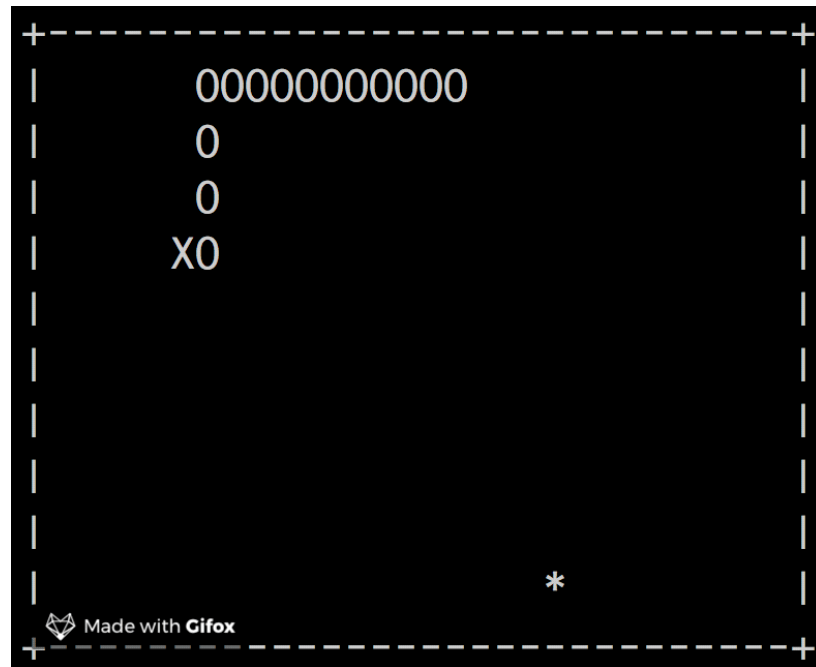
- Why this game?
- Prerequisites
- Different development steps
 - Flow control
 - Variables and constants
 - Predefined functions
 - Arrays
 - Functions
 - I/O

SUMMARY



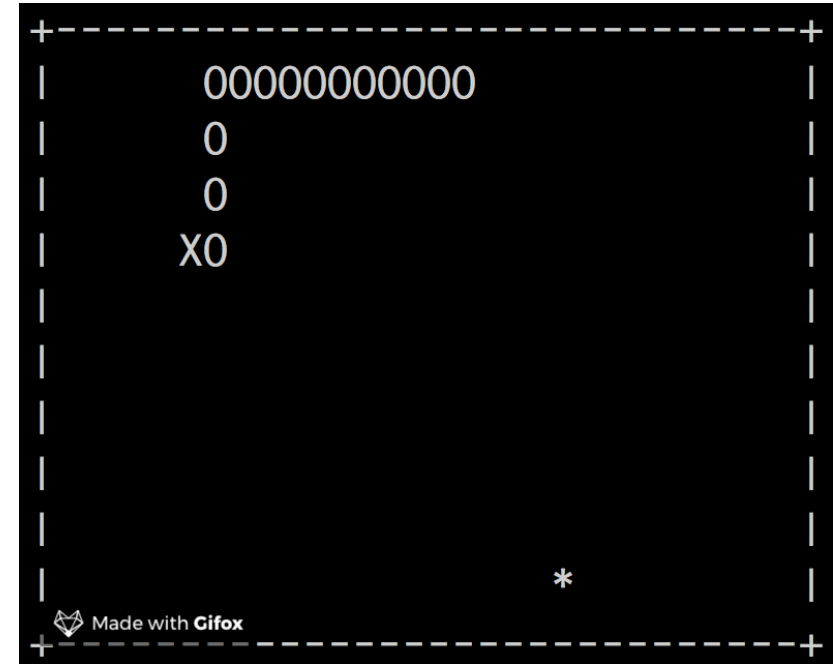
Why this game?

- The idea is giving you a project that will be continuously improved up to the end of this course
- Console based graphic
 - Back to '70s
- Please note that these slides will be continuously updated!



Why this game?

- What is the snake program?
- It is a simple game that can easily be rendered using only the console (1976)
- The player control a “snake” that is continuously moving
- Each time the “head” of the snake “eat” some “food”, the snake itself grows becoming longer and longer
- When the snake hits the “walls” or itself it is a game over!



- Even console “graphics” can be difficult
- 2 main issues:
 - In a game I would like to read keypresses without stopping the program
 - `scanf()` and other console input functions stops until a “return” is entered...
 - I also would like to be able to write in a given position
 - `printf()` and other console output functions can not behave like this...

- We will start from a C “skeleton” that contains specific functions to solve those issues
- Therefore:
 - Create a “snake” project in your folder (U:\ for Lab workstations)
 - Overwrite the main.c in your project with the skel.c in the “snake” folder in the lab repository
 - Try to compile & run

10. Flow control: selection

- Print a menu like:
 1. Start game
 2. Scores
 3. Help
 4. Quit
- Read from keyboard the user choice

- Print the choice (as debug) and act as follows
 - Help:
 - Print “Q: left, E: right, W: up, S: down, ‘space’: pause, X: quit”
 - Quit: end the program
 - Scores: do nothing for now
 - Start: go on to the next slide
- What is the best selection statement to use?

20. Flow Control: cycles

- Modify step #10 in order to print again the menu:
 - When the user enters a wrong choice
 - After the help
 - Later, also after the start...
- When the user enters the “start” go on according to next slides

- Print the game field
- Namely, a **23×97** rectangle surrounded by a border
 - Use '+' for corners
 - Use '-' for horizontal borders
 - Use '|' (ASCII 124) for vertical borders
- Use a top down approach!

- Hints for printing the playing field
 - Do not stick on a given size for the field
 - Put the 23 and 97 in variables → “generalize” your approach
 - Use a top down approach
 - Imagine yourself driving down the field using the limitations of the printf()
 - What we have to write initially? And then? How many times?
- Questions:
 - How many cycles do we need?
 - Do we need to nest some cycles?

30. Variables & Constant

- Use a variable for the score and init it to zero
 - What is the best type to use?
- Modify 20#:
 - Print the score before the playing field using 6 characters and using zero as padding character
 - Use constants for the symbols used for the border of the playing field
 - Use constants for the size of the playing field
- Define also variables for the snake head position and heading
 - Initialize them with random values
 - Which kind of variables do we have to use? How many?

40. Predefined functions: usleep()

- Study the following functions (on the book!)

```
#include <unistd.h>
```

```
int usleep(usec);
```

- It stops the program execution for usec μ s (1μ s = 1000 ns)
- It can be used during the game (later)

40. Predefined functions

- In the skeleton you can find other functions
- They are **NOT** C predefined functions
- But they have been already fully defined
- Then we can use them as predefined functions anyway

40. “Predefined” functions: clearscreen()

`void clearscreen(void)`

- It “clears” the console
- Then it can be used when starting the program, when switching back to the menu after playing, ...

40. “Predefined” functions: gotoxy()

```
void gotoxy(int, int)
```

- When using printf() and similar functions, the output is written in a constrained position in the console
 - “printing cursor” position
 - From left to right, from the top to bottom
- In a game I have to be more free about printing position
- The gotoxy() function allow to move the cursor in a given position
 - row and column coordinates
 - The top-left position in the console has (0, 0) coordinates
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40. “Predefined” functions: getcommand()

char getcommand(void)

- scanf() and similar keyboard input functions “stop” the execution
 - They wait for a “enter/return”
- In a game we need to be able to read keystrokes in a non blocking fashion
- getcommand() returns the ASCII value of a key that has been pushed or 0 if no key has been stroke

50. Start the game!

- In #30 you defined and randomly initialized the variables for storing the snake position and movement
- Exploiting the gotoxy() functions:
 - Write the snake head 'X' in the console!

60. Update the game!

- After printing the playing field & the snake head:
 - Use an infinite loop for continuously updating the console
 - Each cycle:
 - Update the snake head position according to the randomly chosen direction where the snake is heading
 - Check whether the snake hits a border, in such a case exit the infinite loop
- Which instructions I can use for an “infinite” loop?
- How can I exit such a loop?

60. Update the game! (2)

- In #30 you should have defined which kind of variables to use for storing snake head position and its heading
 - For position it is fine to use 2 variables for the 2 coordinates
 - But what is the best way to store the direction where the snake is heading?
- Most of you use a single variable to encode the direction
 - e.g. 1 for heading left, 2 for right etc.

60. Update the game! (2)

- This choice can be effective but it is not so efficient
 - Each time we have to “transcode” the value
- Alternative solution:
 - The heading can be seen as a vector
 - Encode horizontal & vertical components
 - Pros:
 - Movements → directly encoded
 - Change of direction → easy
 - Cons:
 - We need 2 variables

60. Update the game! (3)

- A simple 90° rotation, can be used to update the state when a “turn” is requested

$$\begin{pmatrix} nm_x \\ nm_y \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} m_x \\ m_y \end{pmatrix}$$

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70. read the input

- Use the `char getcommand(void)` function to read the keyboard input
- Move the snake head accordingly:
 - Q: left
 - E: right
 - W: up
 - S: down

80. add some food & poison

- Randomly (i.e. not each cycle) add some food ('#') and some poison ('Y') in a random position (i.e. random column/row) in the playing field
- Increase/Decrease the player score when the snake head get it!
- Just one piece of food and one piece of poison (until the snake eats it!)

90. arrays: add snake body

- When the snake head hits the food:
 - Food must be removed from the playing field
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 - Initially no body
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- For poison implement an opposite behavior
- Use arrays for dealing with the body
 - A single array or multiple ones and what size?
 - Which kind of data I need?
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100. Dynamic memory allocation

- Use a dynamically allocated array to deal with snake body
 - Use `malloc()` to allocate first body piece
 - Use `realloc()` to grow/reduce it piece by piece
 - Use `free()` when the game ends

120. User-defined functions

- Factorize your code using functions, e.g.:
 - menu()
 - printfield()
 - printsnake()
 - putfood()
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 - updatefield()
 - ...
- Barely nothing other than function invocations in the main()!



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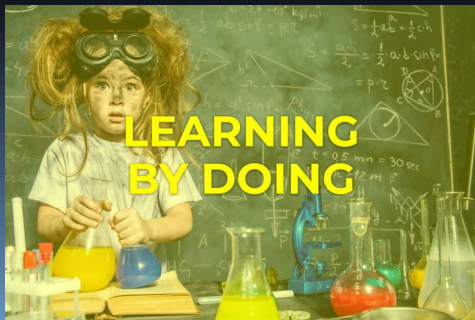
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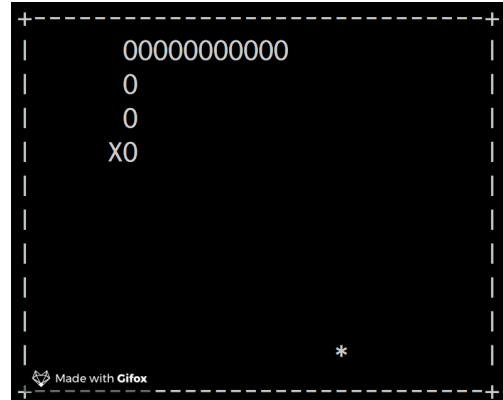


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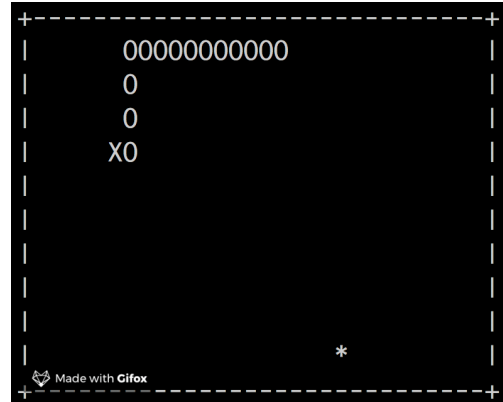
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